

Islamic University of Technology
Department of Electrical and Electronic Engineering

EEE 4700/4800: Project and Thesis

Proposal for Project and Thesis (Capstone Project)

Academic Year		2023-2024	
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1. Tentative title

Machine Learning-Based Prediction and Optimization of Handover Events in Mobile Networks: A Dataset-Driven Approach from Drive Testing

2. Executive summary

This research aims to enhance handover processes in mobile networks using machine learning (ML) techniques. A dataset collected through drive testing will be analyzed to either predict handover events or optimize the number of handovers, improving network efficiency and user experience. By leveraging ML, the study seeks to address current challenges in seamless connectivity, particularly in high-mobility environments, where frequent handovers may degrade performance. The outcome of this research will provide insights into more efficient network management strategies and contribute to advancements in mobile communication technology.

3. Goals of the project

- Analyze drive test data to understand handover patterns in mobile networks.
- Use machine learning to predict handover events or optimize handover frequency.
- Improve network efficiency by reducing unnecessary handovers.
- Enhance user experience by minimizing connection disruptions during handovers.
- Develop a data-driven framework for better handover management in high-mobility environments.

4. Motivation and background

As mobile networks evolve, especially with the expansion of 5G technology, maintaining seamless connectivity during handovers becomes increasingly critical. Handover, the process of transferring an active connection from one cell or base station to another, is essential for uninterrupted service in high-mobility scenarios. However, inefficient handovers can lead to dropped connections, latency, and poor user experience, particularly in dense urban environments or fast-moving vehicles.

Traditional handover management approaches rely heavily on signal strength and predefined thresholds, often resulting in unnecessary or delayed handovers. With the growing complexity of network architectures and user mobility patterns, these conventional methods may not be sufficient. Machine learning offers the potential to revolutionize this process by predicting or optimizing handovers based on real-time data, enabling more intelligent decision-making.

The motivation for this research stems from the need to improve handover efficiency in mobile networks, reduce service interruptions, and enhance overall network performance. By utilizing data collected from drive tests and applying machine learning techniques, this study aims to develop a smarter approach to handover management, contributing to the ongoing advancements in mobile communication technology.

5. Description of project implementation methodology

This project aims to implement machine learning for optimizing handovers in mobile networks, with a focus on reducing unnecessary handovers, preventing failures, and improving overall network performance. The implementation begins with data collection using XCAL, a drive test tool that captures real-time network parameters, which will provide the foundation for machine learning training. The collected data will include signal strength metrics such as RSRP, RSSI, RSRQ, and SINR, along with handover events, timestamps, and GPS data to track the UE's movement. This data is essential for understanding the current handover performance and setting a baseline.

The next step involves feature extraction and preprocessing of the data. Critical features such as signal strengths from serving and neighboring cells, user speed, and the location of handover events will be used to train the machine learning models. Various models will be evaluated, including decision trees, random forests, support vector machines (SVM), Gradient Boosting Machines (GBM), and time-series models like LSTMs, which are well-suited for sequential data. The training process involves labeling the data as "handover required" or "handover not required," allowing the models to learn the conditions under which handovers should be initiated.

To improve the accuracy and robustness of predictions, cross-validation techniques will be employed, ensuring that the models generalize well to unseen data. The performance of these models will be evaluated using metrics such as accuracy, precision, recall, and F1 score. The model that best balances

these metrics while minimizing computational complexity will be selected as the best-fit candidate for deployment.

The deep learning approach will involve training the model on the extracted features from the drive test data, allowing it to learn from historical patterns and improve its predictions over time. The resulting model will be evaluated based on its ability to minimize false positives and negatives during handover predictions, ensuring that only necessary handovers are initiated. This advanced method aims to optimize handover timing, ultimately leading to enhanced network performance and an improved user experience.

Once the optimal model is deployed, it will predict the optimal handover timing based on real-time network conditions. The model will aim to reduce unnecessary handovers by predicting events only when the signal strength of neighboring cells is consistently better than the current serving cell. This strategy will also prevent ping-pong handovers by recognizing patterns of frequent short-term transitions between cells. The impact of the model will be evaluated through a before-and-after comparison, measuring the reduction in handover frequency, improvement in handover success rates, and enhancements in call quality and energy savings.

6. Description of how complex engineering problems can be addressed

Handover optimization in mobile communication networks presents a complex engineering challenge due to the dynamic nature of network conditions, user mobility, and varying signal quality. To address this complexity, a systematic approach involving data-driven methodologies is employed. By collecting real-time data during drive tests using XCAL, we gather comprehensive information on signal strength metrics, user equipment speed, and historical handover events. Analyzing this data through advanced analytics allows us to identify patterns and correlations, which serve as the foundation for predictive modeling. This enables us to anticipate the optimal timing and conditions for handovers, minimizing unnecessary transitions that could disrupt user experience.

Moreover, incorporating machine learning techniques provides a robust framework for decision-making in handover management. Through the training of various machine learning models, we can evaluate their performance in predicting when handovers are necessary. By comparing models based on their accuracy, precision, and recall, we identify the best fit for our specific scenario. This iterative process of model selection and refinement allows us to enhance the efficiency of handover decisions, ultimately resulting in a reduction in failed handovers and improved overall network performance.

Lastly, the importance of stakeholder engagement cannot be overlooked in the optimization process. Collaboration with network operators and engineers ensures that the developed solutions align with practical deployment scenarios and operational constraints. Continuous feedback loops from real-world implementations facilitate the iterative improvement of our models and strategies, allowing us to adapt to changing network conditions and user behaviors. This integrated approach not only addresses the technical challenges of handover optimization but also ensures that solutions are viable and accepted within the operational framework of mobile networks.

7. Socio-cultural, environmental, and ethical impact of the project

Socio-Cultural Impact

The handover optimization project has the potential to significantly enhance the user experience in mobile communications, which is integral to modern social interactions and cultural engagement. By reducing call drop rates and improving overall connectivity, the project supports seamless communication for individuals and communities, fostering stronger social ties and greater access to information. Enhanced mobile services can empower users in remote areas, bridging the digital divide and facilitating social inclusion. Moreover, improved connectivity can bolster local economies by

supporting businesses that rely on mobile communication, enabling them to reach a broader audience and engage in e-commerce.

Environmental Impact

From an environmental perspective, optimizing handover processes can lead to reduced energy consumption in mobile networks. Fewer unnecessary handovers mean that user equipment (UE) spends less time engaging in signaling processes, leading to lower power usage. This not only benefits individual users through longer battery life but also contributes to the overall energy efficiency of network infrastructure. By minimizing the signaling load on base stations, the project can potentially reduce carbon emissions associated with energy consumption in mobile networks. Ultimately, a more efficient network can support sustainability initiatives by promoting eco-friendly practices in the telecommunications sector.

Ethical Impact

The ethical considerations of the handover optimization project revolve around data privacy and algorithmic fairness. As the project involves the collection of real-time user data through drive tests, it is essential to ensure that data handling practices adhere to privacy regulations and ethical standards. Transparency in data usage and obtaining informed consent from users will be prioritized to maintain trust. Additionally, machine learning models used for predicting handovers must be designed to avoid bias, ensuring that all users receive equitable service regardless of their location or usage patterns. Addressing these ethical concerns will not only enhance the project's credibility but also foster responsible innovation in the field of telecommunications.

8. The following table shows the COs for Project and Thesis along with the required coverage in EEE 4700 and EEE 4800. If any of these COs have additional coverage in EEE 4700 or EEE 4800, indicate them by putting tick. Briefly describe how these COs can be achieved.

COs	CO Statement	POs	Put Tick (√)	
			EEE 4700	EEE 4800
CO1	Identify a contemporary real life problem related to electrical and electronic engineering by reviewing and analyzing existing research works.	PO2	√	
CO2	Determine functional requirements of the problem considering feasibility and efficiency through analysis and synthesis of information.	PO4	√	
CO3	Select a suitable solution and determine its method considering professional ethics, codes and standards.	PO8	√	
CO4	Adopt modern engineering resources and tools for the solution of the problem.	PO5	√	
CO5	Prepare management plan and budgetary implications for the solution of the problem.	PO11	√	
CO6	Analyze the impact of the proposed solution on health, safety, culture and society.	PO6	√	
CO7	Analyze the impact of the proposed solution on environment and sustainability.	PO7	√	
CO8	Develop a viable solution considering health, safety, cultural, societal and environmental aspects.	PO3		√
CO9	Work effectively as an individual and as a team member for the accomplishment of the solution.	PO9	√	√

CO10	Prepare various technical reports, design documentation, and deliver effective presentations for demonstration of the solution.	PO10	√	√
CO11	Recognize the need for continuing education and participation in professional societies and meetings.	PO12		√

9. The aspects that can be addressed for the following Program Outcomes (POs) (indicate by putting tick).

	Statement	Different Aspects	Put Tick (√)
PO3	Design/development of solutions: Design solutions for complex electrical and electronic engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	Public health	
		Safety	
		Cultural	
		Societal	√
		Environmental	
PO4	Investigation: Conduct investigations of complex electrical and electronic engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	Design of experiments	√
		Analysis and interpretation of data	√
		Synthesis of information	√
PO6	The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex electrical and electronic engineering problems.	Societal	√
		Health	
		Safety	
		Legal	√
		Cultural	
PO7	Environment and sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex electrical and electronic engineering problems in societal and environmental contexts.	Societal	√
		Environmental	
PO8	Ethics: Apply ethical principles embedded with religious values, professional ethics and responsibilities, and norms of electrical and electronic engineering practice.	Religious values	
		Professional ethics and responsibilities	√
		Norms	√
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	Comprehend and write effective reports	√
		Design documentation	√
		Make effective presentations	√
		Give and receive clear instructions	√
PO11	Project management and finance: Demonstrate knowledge and understanding of engineering management principles and economic	Engineering management principles	√

	decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	Economic decision-making	√
		Manage projects	√
		Multidisciplinary environments	

10. The attributes of Knowledge Profile (K3 – K8) that can be addressed (indicate by putting tick).

K	Knowledge Profile (Attribute)	Put Tick (√)
K3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline	√
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline	√
K5	Knowledge that supports engineering design in a practice area	√
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline	√
K7	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability	
K8	Engagement with selected knowledge in the research literature of the discipline	√

Explanation or justification of how the Knowledge Profiles (K3 – K8) will be addressed in EEE 4700/4800 (Project and Thesis).

K	Explanation/Justification
K3	The systematic, theory-based formulation of engineering fundamentals essential for the engineering discipline directly supports our thesis on handover optimization by providing a solid foundation in telecommunications principles, signal processing, and network design. Understanding these core theories enables the development of advanced algorithms and models that accurately predict and optimize handover processes, ensuring improved network performance and enhanced user experience. By applying these engineering fundamentals, the project aims to address real-world challenges in mobile communications effectively, contributing to the overall advancement of the telecommunications field.
K4	The thesis on handover optimization leverages specialized knowledge in telecommunications engineering, particularly in areas like radio frequency (RF) management, mobility protocols, and machine learning. These theoretical frameworks form the foundation for developing innovative models that enhance handover decision-making processes. By using cutting-edge techniques, the research remains at the forefront of the discipline, advancing best practices in optimizing network performance and reliability in mobile communication systems
K5	In the context of handover optimization, the knowledge supporting engineering design includes understanding signal strength metrics (RSRP, RSSI, RSRQ), mobility management, and machine learning techniques. This foundational knowledge guides the design of predictive models that enhance handover decisions in real-time, ensuring efficient network performance. By applying this knowledge, the design of the optimization process reduces unnecessary handovers and improves user experience, aligning with best practices in mobile network engineering.

K6	In handover optimization, engineering practice knowledge involves the application of technologies such as XCAL for data collection during drive tests and machine learning for predictive analysis. This knowledge includes practical skills in configuring and managing network parameters (e.g., signal strength, TTT, and cell handover events), and leveraging tools for real-time network monitoring and optimization. It ensures the effective implementation of handover solutions in live networks, aligning with industry standards and practices in telecommunications engineering.
K7	
K8	In the context of handover optimization using machine learning, engagement with the research literature involves critically analyzing and applying relevant studies on signal strength metrics, mobility management, and predictive models for handover decisions. By exploring existing frameworks and methodologies, such as machine learning-based optimization techniques, the research leverages state-of-the-art findings to improve network efficiency and user experience. This engagement also includes evaluating advancements in mobility management, network performance, and the use of data-driven approaches to enhance handover timing and reduce unnecessary transitions in mobile networks.

11. The Attributes of Ranges of Complex Engineering Problem Solving (P1 – P7) that can be addressed (indicate by putting tick).

P	Range of Complex Engineering Problem Solving	Put Tick (✓)
Attribute	Complex Engineering Problems have characteristic P1 and some or all of P2 to P7:	
Depth of knowledge required	P1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach	✓
Range of conflicting requirements	P2: Involve wide-ranging or conflicting technical, engineering and other issues	✓
Depth of analysis required	P3: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models	✓
Familiarity of issues	P4: Involve infrequently encountered issues	✓
Extent of applicable codes	P5: Are outside problems encompassed by standards and codes of practice for professional engineering	
Extent of stakeholder involvement and conflicting requirements	P6: Involve diverse groups of stakeholders with widely varying needs	
Interdependence	P7: Are high level problems including many component parts or sub-problems	✓

Explanation or justification of how the attributes of ranges of Complex Engineering Problem Solving (P1 – P7) will be addressed in EEE 4700/4800 (Project and Thesis).

P	Explanation/Justification
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P1	Handover optimization in mobile networks requires in-depth engineering knowledge, particularly in wireless communication principles (K3), signal processing (K4), and predictive modeling (K6). The challenge involves applying first principles and fundamentals, such as radio signal propagation and machine learning algorithms, to resolve complex, data-driven decisions for optimal handover timing.
P2	Conflicting requirements arise in balancing user experience, network performance, and energy efficiency. For example, reducing handovers can conserve resources but might degrade service quality if poorly timed. The optimization needs to address both technical performance and user satisfaction, making it a multifaceted issue.
P3	The problem lacks an obvious solution due to the dynamic nature of mobile networks, which requires abstract thinking and modeling. Creating predictive models for handover decisions demands originality, as real-time factors like user mobility, signal fluctuation, and network congestion need to be accounted for
P4	Handover optimization encounters infrequent issues, especially with the emergence of new technologies like LTE, 5G. Problems such as ping-pong handovers or rare network outages require tailored solutions, often necessitating innovative analysis beyond standard procedures.
P5	
P6	
P7	Handover optimization is a high-level engineering challenge that consists of multiple sub-problems, including signal strength assessment, mobility tracking, and real-time decision-making. Each sub-problem is interdependent, requiring holistic consideration of various factors to ensure network efficiency and user satisfaction.

12. Attributes of Ranges of Complex Engineering Activities (A1 – A5) that can be addressed (indicate by putting tick).

A	Range of Complex Engineering Activities	Put Tick (✓)
Attribute	Complex activities means (engineering) activities or projects that have some or all of the following characteristics:	
Range of resources	A1: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies)	✓
Level of interaction	A2: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues	✓
Innovation	A3: Involve creative use of engineering principles and research-based knowledge in novel ways	✓
Consequences for society and the environment	A4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation	
Familiarity	A5: Can extend beyond previous experiences by applying principles-based approaches	✓

Explanation or justification of how the attributes of ranges of Complex Engineering Activities (A1 – A5) have been addressed in EEE 4700/4800 (Project and Thesis).

A	Explanation/Justification
A1	The thesis project on handover optimization utilizes diverse resources such as data collection tools (XCAL), machine learning technologies, financial budgeting for drive tests, and collaboration with network engineers. It also involves managing large-scale datasets and computational resources for model training and evaluation.
A2	The project addresses significant problems, such as conflicting requirements between minimizing handovers and maintaining service quality. These technical challenges require balancing network performance, user experience, and operational efficiency, making it a multi-dimensional task with intricate interactions.

A3	The project applies machine learning in novel ways to optimize handover processes, which is a creative use of engineering principles. By integrating real-time network data and predictive algorithms, the research develops innovative solutions beyond conventional handover strategies used in mobile networks.
A4	
A5	This project extends beyond previous experiences by applying principles of signal processing and machine learning to the evolving challenges of mobile networks. The methods used build upon established concepts but adapt them to new, real-world challenges in 4G and beyond.

13. Work plan of the project

Winter Semester	Data Collection: Drive tests using XCAL are to be conducted to gather network data (RSRP, RSSI, RSRQ, SINR, GPS, handover events). Preprocessing and cleaning of collected data will be done to prepare it for machine learning model training. Relevant features such as signal strength metrics and handover events are to be extracted for further analysis. Initial Model Development: Key performance metrics for handover optimization (accuracy, precision, recall, handover success rate) will be defined. Machine learning models such as Random Forest and Gradient Boosting are to be trained on the dataset. Cross-validation and hyperparameter tuning will be applied to optimize model performance. Preliminary Evaluation: Preliminary evaluations will be conducted to assess handover frequency reduction and success rates. Initial results are to be documented, and areas for model improvement will be identified.
Summer Semester	Deep Learning Implementation: Deep learning models will be developed to improve handover prediction, using sequential data like historical handover patterns. Advanced features such as time-to-trigger and user mobility patterns will be integrated for further optimization. Comprehensive Testing: New drive test data will be used to test model performance in real-world conditions. Comparative analysis of machine learning and deep learning models will be conducted, focusing on handover reduction, success rate, and user experience. Final Evaluation and Reporting: Final evaluation of the model's performance on key metrics (handover frequency, false positives/negatives, call quality, energy savings) will be completed. The thesis will be written, summarizing the results, methodologies, and the overall impact on network optimization.

14. Proposed budget for implementing the project

Sl	Items	Justification	Price (BDT)
1.	Transportation	For the drive test we need to use transportation	7000
2.	Mobile Charge	Sometimes we need to stay in call for several minutes	1000
3.	Miscellaneous Cost	There are miscellaneous costs like stationary items etc.	1500
Total cost (BDT):9,500 BDT			
In words: Nine Thousand Five Hundred taka only			

Student's Signature		Date	

Supervisor's Signature		Date	
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